

1. General Description

VD1405-FOH is a reflective electrophoretic E Ink® technology display module with E-Ink Carta1300 with FastGLR and Eclipse mode based on active matrix

TFT substrate. It has 6" active area with 1072 x 1448 pixels, the display is capable to display images at 2-16 gray levels (1-4 bits) depending on the display controller and the associated waveform file it used.

2. Features

- Carta1300 high contrast reflective/electrophoretic technology
- Fast Regal waveform refresh
- FastGLR
- Eclipse Mode
- 1072 x 1448 resolution
- Ultra wide viewing angle
- Low power consumption
- Pure reflective mode
- Bi-stable
- Commercial temperature range
- Landscape, portrait mode
- Color Temperature Modulation Front Light with white and warm color
- Capacitive Glass Touch w/ support two finger

3. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	6" (3 : 4)	Inch	
Display Resolution	1072 (H) × 1448(V)	Pixel	
Active Area	90.584 (H) × 122.356 (V)	mm	
Pixel Pitch	0.0845 (H) × 0.0845 (V)	mm	
Pixel Configuration	Square		
Outline Dimension	101.8(H) x 138.4 (V) x 1.94(D) (EPD w/o FPC, TFT is 0.4t)	mm	
Module Weight	47.9±2	g	
Number of Gray	16 Gray Level (monochrome)		
Display operating mode	Reflective mode		
Surface treatment	haze 11% +/-3%, hardness >=2H"		2H Hardness, module only

Surface Protective Film

Model Name	Peeling Force (Adhesion to AG)
YHM protective film, 6-inch	25 ± 15 (gf/25mm),

Front View

Dimensions (mm):

- 101.80±0.2(TFT)
- 99.40±0.2(GTP)
- 98.40±0.2(LGP)
- 97.20±0.2(AG)
- 90.584±0.2(AA)
- 4.006
- 89.35±1
- 138.40±0.2(TFT)
- 135.22±0.2(LGP)
- 131.40±0.2(GTP)
- 127.10±0.2(AG)
- 122.356±0.2(AA)
- 22.80±1
- 5.85
- 5.608
- 30.07±1
- 22.17±1
- 1.5±0.3
- 16.5±0.15
- 0.5±0.1
- 0.3±0.03
- 0.5±0.05
- 0.3±0.05 (Stiffener PI)

Bottom View

Dimensions (mm):

- 2.033±0.2(AG+TP)
- 1.938±0.2
- 3.827±1
- 27.070±1
- 4.5±0.07
- 3.5±0.07
- 0.5±0.1
- 0.5±0.03
- 0.3±0.03
- 0.3±0.03 (Gold Finger)
- 2.5±0.3
- 8
- 1

DETAIL A SCALE 2:1

DETAIL B Scale 4:1

DETAIL C SCALE 5:1

NOTES:

- General Tolerance is ±0.3 mm.
- Dimension in () is for reference.
- Label thickness on EPD FPC is 0.06mm.

1.2(Max Glass Bottom to FPC folding height can be flat not dead-folding)

3.55 max(Mouldure + FPC folding height)

Rev

Rev	Description	Design	Date
01	Initial drawing	Chenry	2023/05/22

SCALE

SCALE	UNIT
1:1	mm

APPROVE

APPROVE	CH D
CHECK	CH D
DESIGN	Chenry

PROJECTION METHOD DESCRIPTION

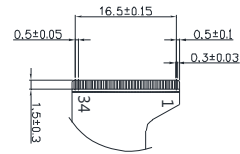
ED060KH-E

Module Drawing

1/1

FILE VERSION : 2023/05/24

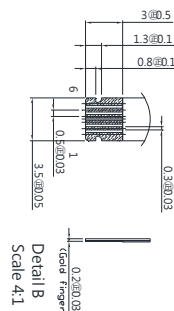
REV.	DESCRIPTION	DESIGN	DATE
01	Initial drawing	Chenry	20230822



DETAIL A
SCALE 2:1

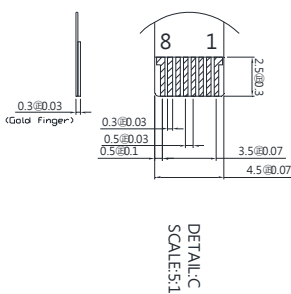
EPD FPC PIN assignment			Description
PIN	Signal	I/O	
1	VSL	P	Negative power supply source driver
2	VCL	P	Negative power supply gate driver
3	VSS	P	Ground
4	NC	-	No Connection
5	NC	-	No Connection
6	VDD	P	Digital power supply drivers
7	VSS	P	Ground
8	CKH	I	Clock source driver
9	VSS	P	Ground
10	LEH	I	Latch enable source driver
11	DEH	I	Output enable source driver
12	SPH	I	Start pulse source driver
13	D0	I	Data signal source driver(LSB)
14	D1	I	Data signal source driver
15	D2	I	Data signal source driver
16	D3	I	Data signal source driver
17	D4	I	Data signal source driver
18	D5	I	Data signal source driver
19	D6	I	Data signal source driver
20	D7	I	Data signal source driver (MSB)
21	VCOM	P	Common voltage
22	NC	-	No Connection
23	NC	-	No Connection
24	NC	-	No Connection
25	XDN	-	XDN control for night mode (default pull high)
26	VSS	P	Ground
27	DEV	I	Output mode selection gate driver
28	CKV	I	Clock gate driver
29	SPV	I	Start pulse gate driver
30	NC	-	No Connection
31	Border	I	Border connection
32	VSS	P	Ground
33	VSH	P	Positive power supply source driver
34	VGH	P	Positive power supply gate driver

FLPIN assignment	
PIN	Pin assignment
1	Anode(Circuit01)
2	Anode(Circuit01)
3	Anode(Circuit02)
4	Cathode(Circuit02)
5	Cathode(Circuit01)
6	Cathode(Circuit01)



DETAIL B
Scale 4:1

TP PIN assignment	
PIN	Symbol
1	VSS
2	TPDD_TOUCH_1V8
3	TPDD_TOUCH_RESET_B
4	TPDD_TOUCH_RESET_A
5	TPDD_TOUCH_I2C_SDA
6	TPDD_TOUCH_I2C_SCL
7	TPDD_TOUCH_SWDD
8	VDDA_TP



DETAIL C
SCALE 5:1

NOTES:

1. General Tolerance is ± 0.3 mm.
2. Dimension in () is for reference.
3. Label thickness on EPD FPC is 0.06mm.

SCALE	UNIT	E Ink 元太科技工業股份有限公司	
1 : 1	mm	PRODUCTION HEAD DESCRIPTION	ED060KHE
APPROVE	CH D	CHECK	CH D
CHECK	CH D	PART NO.	Module Drawing
DESIGN	Chenry	FILE VERSION	REV1/2/3/4
		SHEET	1/1

5. Input/Output Interface

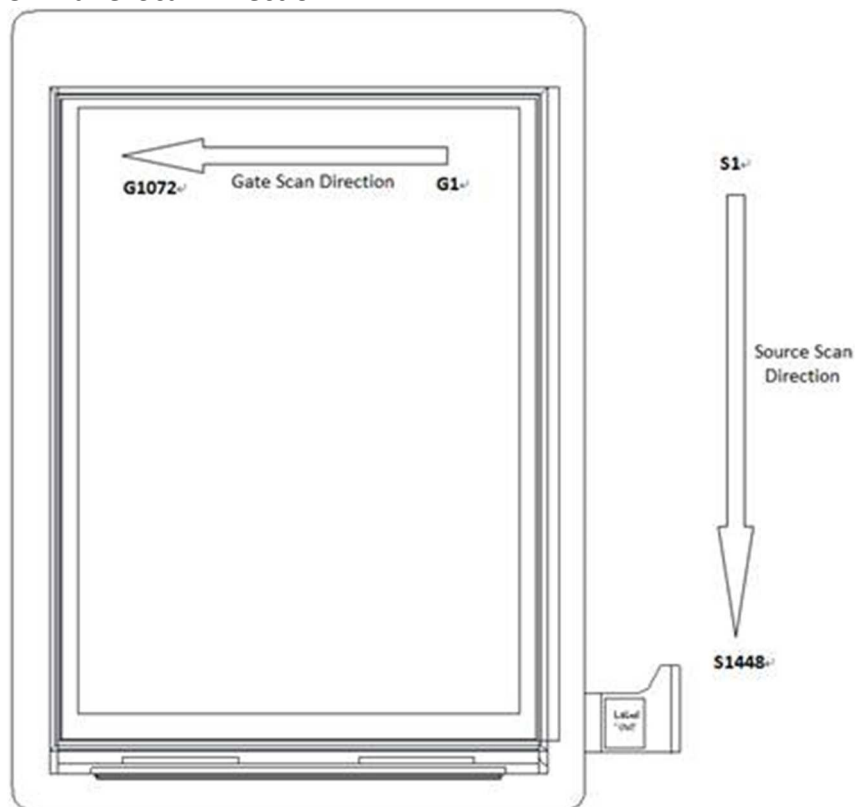
5.1 Pin Assignment of Panel

Connector type : FH34S-34S-0.5SH(50)

Pin #	Signal	I/O	Description	Remark
1	VSL	P	Negative power supply source driver	Note1
2	VGL	P	Negative power supply gate driver	Note1
3	VSS	P	Ground	Note1
4	NC	-	No Connection	
5	NC	-	No Connection	
6	VDD	P	Digital power supply drivers	Note1
7	VSS	P	Ground	Note1
8	CKH	I	Clock source driver	
9	VSS	P	Ground	Note1
10	LEH	I	Latch enable source driver	
11	OEH	I	Output enable source driver	
12	SPH	I	Start pulse source driver	
13	D0	I	Data signal source driver(LSB)	
14	D1	I	Data signal source driver	
15	D2	I	Data signal source driver	
16	D3	I	Data signal source driver	
17	D4	I	Data signal source driver	
18	D5	I	Data signal source driver	
19	D6	I	Data signal source driver	
20	D7	I	Data signal source driver(MSB)	
21	VCOM	P	Common voltage	Note1
22	NC	-	No Connection	
23	NC	-	No Connection	
24	NC	-	No Connection	
25	XON	-	XON control for Eclipse mode (default pull high)	
26	VSS	P	Ground	Note1
27	OEV	I	Output mode selection gate driver	
28	CKV	I	Clock gate driver	
29	SPV	I	Start pulse gate driver	
30	NC	-	No Connection	
31	Border	I	Border connection	
32	VSS	P	Ground	Note1
33	VSH	P	Positive power supply source driver	Note1
34	VGH	P	Positive power supply gate driver	Note1

Note1: P → Power pin

5.2 Panel Scan Direction



5.3 the relationship of input data and output

Output	S1	S2	S3	S4
Data	D7	D5	D3	D1
	D6	D4	D2	D0

5.4 Recommended Connector Type of Panel

FH34S-34S-0.5SH(50)

5.5 Front Light LBR pin assignment

No.	Pin assignment	Remark
1	Cool 1+	Cool 1 (7pcs LED)
2	Cool 1+	
3	Warm 1+	Warm 1 (6pcs LED)
4	Warm 1-	
5	Cool 1-	Cool 1 (7pcs LED)
6	Cool 1-	

5.6 Recommended Connector Type of Touch Panel

FH34SRJ-10S-0.5SH(50)

5.7 Pin Assignment of Touch Panel for Proto only (Note.)

Pin#	Signal	I/O	Description	Remark
1	GND	P	Ground	Note
2	VDD_TOUCH_1V8	P	1.8V volts to touch IC	V _{DD}
3	XRES	I	Reset pin from mainboard to IC 1.8V digital I/O	Request pull high resistor 10K ohm for Reset by Host
4	INT	I/O	Interrupt pin from IC to mainboard 1.8V digital I/O	Pre-layout pull high resistor pad for INT by Host
5	SDA	I/O	I2C data pin to Parade touch IC 1.8V digital I/O	Request pull high resistor 4.7K ohm for SDA by Host
6	SCL	I/O	I2C clock pin to Parade touch IC 1.8V digital I/O	Request pull high resistor 4.7K ohm for SCL by Host
7	NC	-	No Connection	
8	V _{DDA} _TP	P	Power Supply DC 2.7~3.5V	V _{DDA}

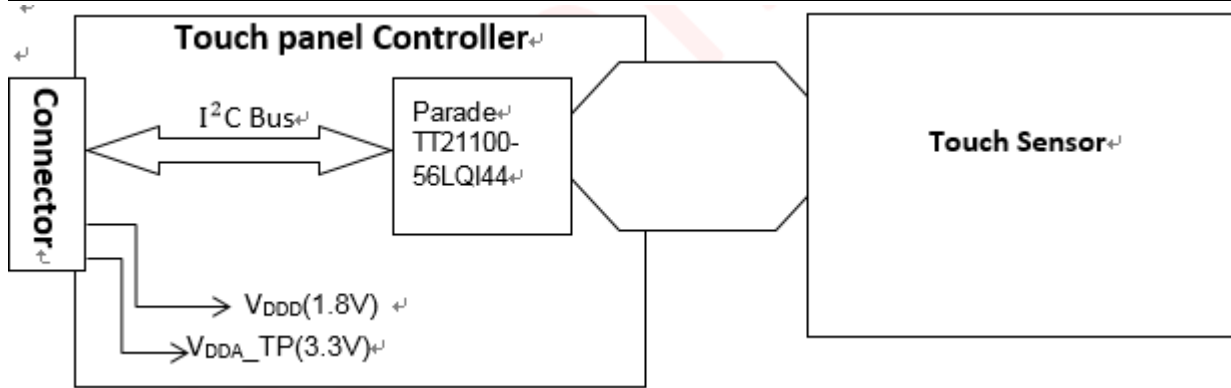
Note: P → Power pin

5.8 Touch Panel Characteristics

5.8.1 Touch Panel Performance by finger (Report Rate: 65Hz)

Symbol		Specification	Remark
Diameter finger		7mm	
Linearity	Center	±1mm	No display refresh
	Border	±2mm	
Accuracy	Center	±1mm	No display refresh
	Border	±2mm	
Jitter	Center	±1mm	No display refresh
	Border	±2mm	

5.8.2 Touch Panel circuit application



5.8.3 Touch driver IC DC Characteristics : (Note.)

Symbol	Description	Conditions	Min	Typ	Max	Units
V _{DDD}	Digital supply voltage	Core LDO is enabled (V _{CCD} output)	2	-	5.5	V
		Core LDO is disabled (V _{CCD} input)	1.71	1.8	1.95	V
V _{CCD}	Digital core supply voltage	Core LDO is enabled (V _{CCD} output)	-	1.8	-	V
		Core LDO is disabled (V _{CCD} input)	1.71	1.8	1.95	V
V _{DDA_TP}	Analog supply voltage	TX pump is enabled	2.65	-	4.7	V
V _{CCTX}	V _{CCTX} Supply operating voltage	Input to external low pass filter. TX pump disabled configuratoin	2.65	-	4.7	V
V _{TX}	XY Pin in Tx output mode	TX pump is enabled	4.75	-	5.25	V
I _{DDD_ACT}	V _{DDD} Active Current	-	-	20	50	mA
I _{DDA_ACT}	V _{DDA} Active Current	-	-	15	18	mA
I _{DDD_DS}	V _{DDD} Deep Sleep Current	-	-	3	-	uA
I _{DDA_DS}	V _{DDA} Deep Sleep Current	-	-	2	-	uA
I _{DDD_XR}	V _{DDD} Current, RESET=LOW	1.71V ≤ V _{DDD} ≤ 1.95V	-	5	-	uA
		2.0V ≤ V _{DDD} ≤ 5.5V	-	1	-	uA
I _{DDA_XR}	V _{DDA} Current, RESET=LOW	-	-	25	-	uA
I _{DDD_P}	V _{DDD} Flash programming and Flash verify current	-	-	5	25	mA

Note:

1. The final touch module consumption will be come out base on device firmware tune lockdown setting.
2. The power consumption is only for reference.
3. In every mode, the actual current will vary due to the number of channels and firmware configuration.

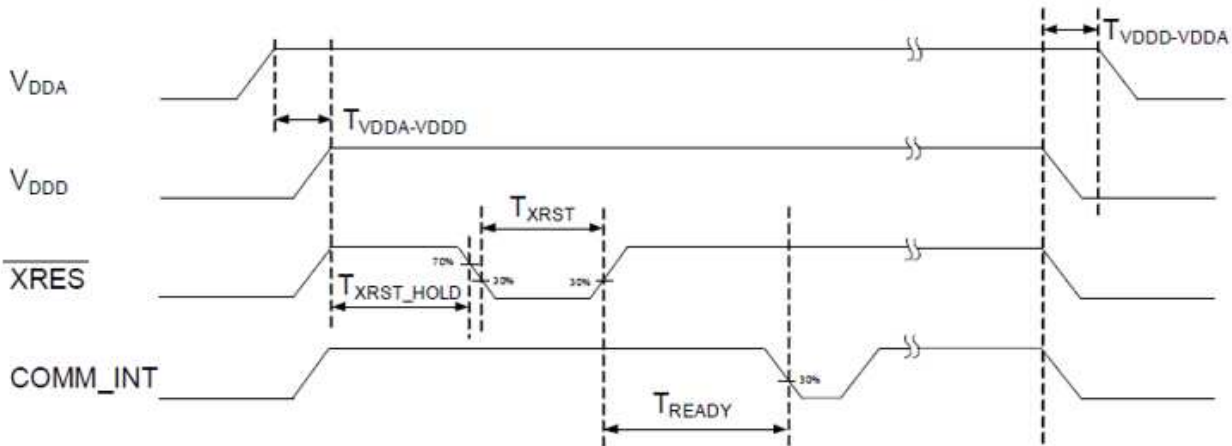
5.8.4 Touch driver IC AC Characteristics and Power Sequence: (Note.)

Symbol	Description	Conditions	Min	Typ	Max	Units
PSA _{RAMP}	V _{DDA} Ramp Up	-	-	-	100	mV/us
PSD _{RAMP}	V _{DDD} Ramp rate from ground to minimum voltage	-	1	-	40	V/ms
T _{VDDA-VDDD}	Time of V _{DDA} Ready time before V _{DDD} ready time	-	0	-	-	ms
T _{XRST_HOLD}	Hold time berfore external reset(XRES) asserted	-	2	-	-	ms
T _{XRST}	External reset(XRES) pulse width	After V _{DDD} is valid	10	-	-	us
T _{VDDD-VDDA}	Time of V _{DDD} discharge before V _{DDA} discharge time	-	0	-	-	ms
T _{READY}	Time from XRES deassertion to COMM_INT	-	-	-	10	ms
T _{CAL}	Calibration routine execution time	-	-	-	1100	ms

Note : V_{DDA_TP} = V_{DDA} = 3.3V

V_{DDD} : Generated inside touch driver based on V_{DDA}

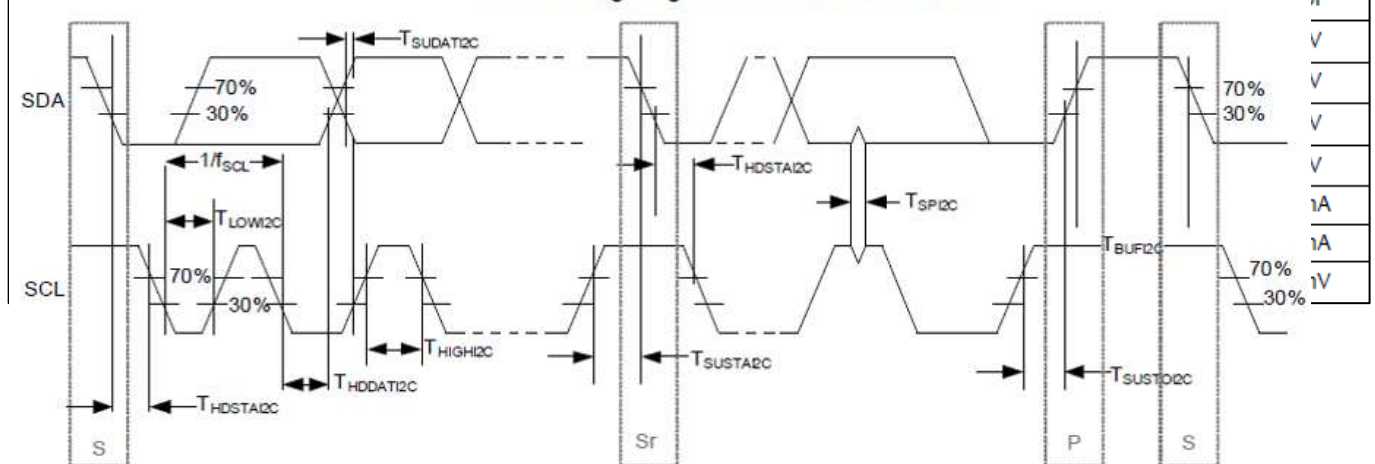
Power Sequence Timing Diagram



5.8.5 Touch driver IC I2C Bus Timing Diagram for Fast/Standard Mode

Symbol	Description	Standard Mode		Fast Mode		Units
		Min	Max	Min	Max	
f_{SCL2C}	SCL Frequency	0	100	0	400	kHz
$T_{HDSTA2C}$	Hold Time (Repeated) Start Condition First Clock Pulse Is Generated after this Period	4	–	0.6	–	μs
T_{LOWI2C}	SCL LOW Period	4.7	–	1.3	–	μs
$T_{HIGHI2C}$	SCL HIGH Period	4	–	0.6	–	μs
$T_{SUSTA2C}$	Setup Time for Repeated Start Condition	4.7	–	0.6	–	μs
$T_{HDDAT2C}$	Data Hold Time	0	–	0	–	μs
$T_{SUDAT2C}$	Data Setup Time	250	–	100	–	ns
$T_{VDDAT2C}$	Data Valid Time	–	3.45	–	0.9	μs
$T_{VDACKI2C}$	Data Acknowledge Time	–	3.45	–	0.9	μs
$T_{SUSTOI2C}$	Setup Time for Stop Condition	4	–	0.6	–	μs
V_{HH}	Input Hysteresis High Voltage, $1.71V \leq V_{DD} \leq 1.95V$ or $2.0V \leq V_{DD} \leq 5.5V$	$0.1 \times V_{DD}$	–	$0.1 \times V_{DD}$	–	V
T_{BUFI2C}	Bus Free Time between a Stop and Start Condition	4.7	–	1.3	–	μs
T_{SPI2C}	Pulse Width of Spikes that are Suppressed by Input Filter	–	–	50	–	ns

I²C Bus Timing Diagram for Fast/Standard Mode



Legend
S: I²C Start Condition
Sr: I²C Repeat Start Condition
P: I²C Stop Condition

5.8.6 Touch Packet Interface Protocol (PIP) Communication

```
Code 1 - Register Pointer Example Write Code
w 24 04 00 05 00 2F 00 03
; Packet Length = 0x0050
; Packet ID = 0x2F
```

```
Code 2 - Register Pointer Example Read Code
w 24 01 00
r 24 x x x
```

Write Command Break-down

Write	I ² C Address	Buffer Offset LSB	Buffer Offset MSB	Packet Length LSB	Packet Length MSB	Report ID	Data[0]	Data[1]
W	24	04	00	05	00	2F	00	03